

SmartShopper



Product Case Study

Building a Grocery Basket Optimization MVP

Helping users reduce grocery spending through centralized real-time price comparison.

By,

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Identifying the Problem

Users currently :

- Compare prices manually across multiple store apps
- Struggle to identify the cheapest overall basket
- Spend significant time evaluating stores individually

Core Problem: Existing grocery platforms optimize for delivery convenience - not total savings.

★ **Insight :** Users optimize grocery shopping based on savings, convenience, and time efficiency - not price alone



Why This Problem Matters?

User Impact : Rising grocery prices are increasing financial pressure on everyday shoppers.

Current comparison workflows create:

- decision fatigue
- repetitive manual effort
- unnecessary app switching

Market Gap : Most grocery platforms:

- optimize for delivery convenience
- focus on individual item pricing
- lack basket-level intelligence

Result : Users still struggle to identify the most cost-effective overall shopping option.

★ **Key Product Insight :** Users optimize grocery shopping based on - total basket savings, convenience, time efficiency - not simply the cheapest individual item.

User Personas



Understanding Our Users : Patterns from interviews revealed three distinct personas - all struggling with the same core problem.

1. Budget-Conscious Students

Who They Are : Fixed-budget student balancing savings and convenience.



Shopping Behavior :

- Compares prices across multiple apps
- Plans weekly shopping around a strict budget
- Looks for the best value on every trip

Biggest Frustrations :

- Manual app-switching takes too long
- Hard to tell which store is cheapest overall
- Price changes create uncertainty

Primary Goals :

- Stay within budget
- Find affordable options quickly
- Spend less time comparing stores



2. Busy Household Shopper

Who They Are : Manages recurring household purchases and values efficiency.

Shopping Behavior :

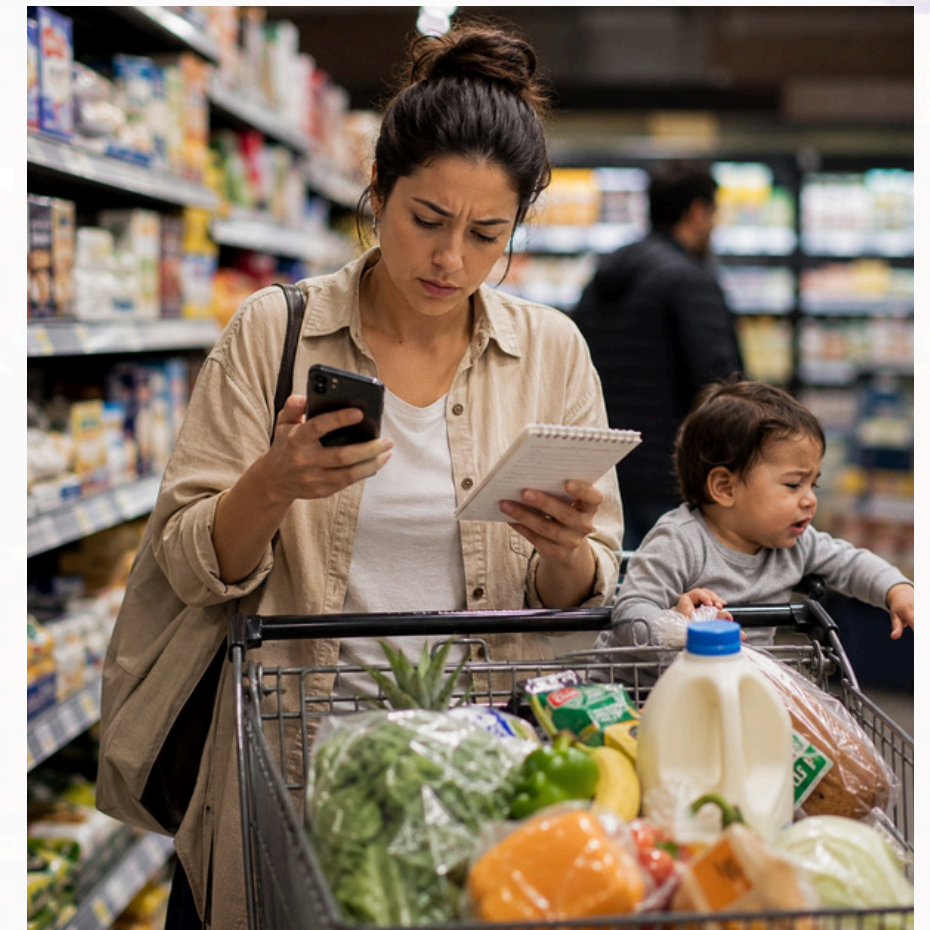
- Buys the same core items each week
- Prioritizes one-trip convenience
- Coordinates around family needs

Biggest Frustrations :

- Comparing stores adds effort
 - Multiple apps feel fragmented
- Shopping time competes with other responsibilities

Primary Goals :

- Complete shopping in fewer steps
- Save time without losing savings
- Keep spending predictable



3. Smart Saver / Deal Optimizer



Who They Are : Value-focused shopper hunting for the best deals.

Shopping Behavior :

- Checks discounts across stores
- Compares item prices before buying
- Switches retailers to maximize savings

Biggest Frustrations :

- Deals are scattered across platforms
- No clear view of total basket savings
- Pricing data is hard to compare quickly

Primary Goals :

- Find the cheapest basket
- See pricing in one place
- Make confident savings-driven decisions



Product Hypothesis & Research



Product Hypothesis

- Reduce shopping time
- Save more money
- Return for repeat usage

Research Participants

- Graduate students
- Recent graduates
- Budget-conscious shoppers

Research Goals

- Understand shopping decision patterns
- Identify frustrations with existing tools
- Validate demand for basket-level comparison
- Discover feature priorities

Success Criteria

- Users complete basket comparisons
- Users return repeatedly
- Users experience measurable savings

MVP Scope



Defining the MVP : The goal was not a fully scaled platform - it was to validate whether users found value in centralized basket optimization.

Included in MVP

- Grocery item search
- Basket creation
- Basket-level price comparison
- Cheapest store recommendation
- Saved shopping lists

Deliberately Deferred

- AI recommendations
- Delivery integrations
- Coupon systems & loyalty programs
- Advanced personalization
- Retailer partnerships

Key decision : Solve one high-impact problem exceptionally well before expanding.

Why These Features Were Prioritized?

The MVP prioritized features that directly addressed the strongest recurring pain points identified during research: repetitive manual comparison, lack of basket-level visibility, and inefficient grocery decision-making. Features that did not directly validate the core hypothesis were intentionally deferred.

Product Strategy & Tradeoffs



Every decision balanced user value, complexity, and validation speed.

- **Web MVP First** : Faster development and iteration without app store constraints.
- **Basket-Level Over Item-Level** : Higher user value identified through research.
- **Simplified UI** : Reduced onboarding friction and cognitive load.
- **Limited Store Integrations** : Reduced technical complexity to accelerate validation.
- **Focused on Single-Store Optimization First** : Users valued convenience alongside savings.

i Key PM Lesson: Strong product management means deciding what not to build initially.

User Experience



The SmartShopper experience was designed to simplify grocery decision-making by reducing manual comparison effort and helping users identify the best overall shopping option quickly.

Rather than optimizing individual item discounts, the experience focuses on:

- basket-level savings
- convenience
- fast decision-making
- low-friction workflows

Core UX Principle: Reduce decision fatigue by simplifying grocery comparison into a single guided workflow.

User Journey



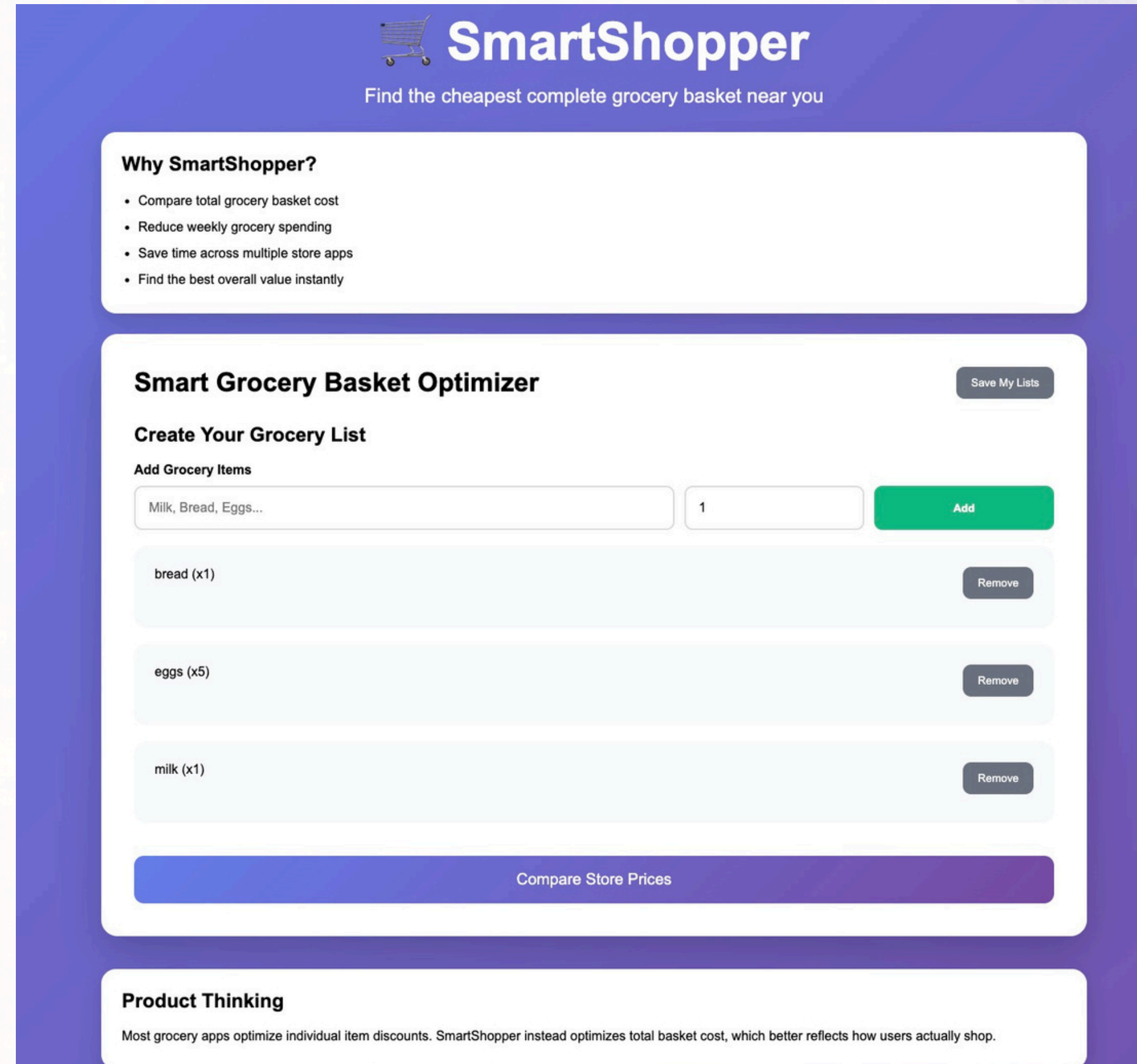
1. Build Grocery Basket: Users begin by creating a shopping basket through a simple and familiar workflow.

Experience Goals:

- Fast item entry
- Minimal learning curve
- Clear basket visibility
- Low cognitive load

Key Product Decisions :

- Simplified basket creation flow
- One-click comparison CTA
- Saved shopping lists for recurring use



The image shows a mockup of the SmartShopper app interface. At the top, there's a blue header with a shopping cart icon and the text "SmartShopper" and "Find the cheapest complete grocery basket near you". Below this, there's a white box titled "Why SmartShopper?" with a bulleted list of benefits: "Compare total grocery basket cost", "Reduce weekly grocery spending", "Save time across multiple store apps", and "Find the best overall value instantly". The main section is titled "Smart Grocery Basket Optimizer" and includes a "Save My Lists" button. Underneath, it says "Create Your Grocery List" and "Add Grocery Items". There's a text input field with "Milk, Bread, Eggs...", a quantity input field with "1", and a green "Add" button. Below this, there's a list of items: "bread (x1)", "eggs (x5)", and "milk (x1)", each with a "Remove" button. At the bottom of this section is a large blue button labeled "Compare Store Prices". The footer section is titled "Product Thinking" and contains the text: "Most grocery apps optimize individual item discounts. SmartShopper instead optimizes total basket cost, which better reflects how users actually shop."

SmartShopper
Find the cheapest complete grocery basket near you

Why SmartShopper?

- Compare total grocery basket cost
- Reduce weekly grocery spending
- Save time across multiple store apps
- Find the best overall value instantly

Smart Grocery Basket Optimizer [Save My Lists](#)

Create Your Grocery List

Add Grocery Items

Milk, Bread, Eggs... 1 [Add](#)

bread (x1) [Remove](#)

eggs (x5) [Remove](#)

milk (x1) [Remove](#)

[Compare Store Prices](#)

Product Thinking

Most grocery apps optimize individual item discounts. SmartShopper instead optimizes total basket cost, which better reflects how users actually shop.

2. Compare Store Prices :

Once the basket is created, SmartShopper aggregates pricing across nearby grocery stores and calculates total basket costs automatically.

Instead of forcing users to compare products individually, the platform surfaces the cheapest overall basket instantly.

Experience Goals :

- Reduce repetitive comparison effort
- Improve pricing transparency
- Enable faster shopping decisions

Key Product Decisions :

- Basket-level optimization prioritized over item-level comparisons
- Store rankings displayed clearly
- Convenience indicators included alongside pricing

Best Store Options

Back



💰 Save \$12.60 by shopping at Walmart

Walmart

\$18.85

BEST VALUE

3.2 miles away

★ Lowest total basket cost

eggs (x3)	\$4.29
bread (x1)	\$2.49
milk (x1)	\$3.49

Trader Joe's

\$18.95

5.1 miles away

★ Good balance of price and convenience

eggs (x3)	\$3.99
bread (x1)	\$2.99
milk (x1)	\$3.99

Stop & Shop

\$19.15

1.9 miles away

★ Closest store to your location

eggs (x3)	\$4.19
bread (x1)	\$2.69
milk (x1)	\$3.89

Target

\$20.05

4.5 miles away

★ Good balance of price and convenience

eggs (x3)	\$4.49
bread (x1)	\$2.79
milk (x1)	\$3.79

Whole Foods

\$31.45

2.8 miles away

★ Closest store to your location

eggs (x3)	\$6.99
bread (x1)	\$4.49
milk (x1)	\$5.99

Continue to Store

3. Highlight Best Shopping Option

The platform identifies the most cost-effective overall shopping option while balancing savings and convenience.

Rather than encouraging multi-store optimization, SmartShopper recommends the best single-store basket to reduce shopping complexity.

Experience Goals :

- Simplify decision-making
- Increase user confidence
- Reduce decision fatigue

Key Product Decisions :

- Best-value recommendation highlighted visually
- Savings surfaced immediately
- Clear pricing hierarchy for faster scanning



Best Grocery Option Found!

SmartShopper found the lowest-cost grocery basket for your shopping trip.

Savings Optimized Successfully 

Start New Comparison

Validation



MVP Validation Goals

The MVP was designed to validate whether users found value in centralized grocery basket optimization.

Primary Validation Questions

- Will users complete basket comparisons?
- Do users value basket-level optimization over item-level comparisons?
- Will users return for repeat usage?
- Does the platform reduce shopping decision effort?

Success Indicators

- Repeat basket comparisons
- Returning users
- Measurable savings
- Saved list usage

The MVP focused on validating user behavior before investing in advanced functionality or retailer-scale integrations.

Metrics



Measuring What Actually Matters

 North Star Metric

Average Savings Per Completed Basket - directly reflects the core value proposition.

Acquisition

Website visits
Signups

Engagement

Baskets created
Repeat comparisons

Retention

Returning users
Weekly active sessions

Outcome

Money saved
Time saved

Initial MVP Targets

50 Active Users	70% Basket Completion Rate
40% Repeat Usage within 7 Days	8 - 12% Estimated Savings Per Basket

Key Takeaways



Through SmartShopper, I learned how to :

- Conduct user-centered research
- Prioritize MVP scope effectively
- Translate insights into product decisions
- Balance UX with technical feasibility
- Define measurable product outcomes
- Work backward from user problems
- Prioritize under constraints
- Make research-driven product decisions

Most Important Takeaway:

Product management is not just about building features - it is about solving meaningful user problems with focused, measurable solutions

Live Demo: [SmartShopper Live Demo](#)